

Name: _____



Science Practice Examination 2024
NCEA Level 1

Achievement Standard 92047
Physics, Earth & Space Science 1.4

Energy in a Physical System
Credits: 5

Suggested Time: 60 minutes

Instructions

- Answer all questions in the spaces provided.
- You must hand this examination paper to the supervisor at the end of the examination.
- Check that this paper has all 9 pages numbered and in the correct order.

You may find the following formulae and values useful:

$$\Delta E = Pt \quad E_k = \frac{1}{2} mv^2 \quad E_p = mg\Delta h \quad W = Fd$$

$$Q = mc\Delta T \quad Q = mL$$

$$P = VI \quad V = IR$$

The specific heat capacity of dry air is $1,006 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$

The specific heat capacity of water vapour is $1,800 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$

Latent heat of vaporisation for water is $2,300,000 \text{ J kg}^{-1}$

Question 1: Mechanical energy

Sarah is camping and there is a zipline set up for her to use.

- a) Sarah (mass=60kg) decides to climb up the ladder to get to the start of the zipline. The ladder is 6.5m above the ground. Calculate how much potential energy she gains when she climbs to the top.

- b) Describe the energy changes as Sarah climbs from the ground to the start of the zipline.

- c) Sarah rides the zipline all the way down to the end. Calculate how fast she will be travelling immediately before the end of the zipline.

State any assumptions made.



d) At the end of her zipline run Sarah was travelling at 8ms^{-1} .

- Explain why this is a different value to the value calculated in part (c).

e) Sarah walks back up the slope to return the zipline to the next person. She notices that it is easier to walk up the slope (250m) than to vertically climb the ladder (6.5m).



Assuming Sarah is able to walk up the slope to the start of the zipline, show (using calculations) that it is 'easier' in terms of force required by Sarah.

Question 2: Thermal energy

The specific heat capacity of dry air is $1,006 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$

The specific heat capacity of water vapour is $1,800 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$

Latent heat of vaporisation for water is $2,300,000 \text{ J kg}^{-1}$

Sarah sets up her tent in a forest clearing. As night becomes dawn, she notices water droplets forming inside her tent while the exterior remains dry. Sarah wonders whether the temperature difference between the warm interior of her tent and the cooler outside air might be causing this phenomenon.



- a) Using the 'particle theory of matter', explain the difference between water droplets and water vapour.

- b) Calculate how much heat must be removed from 10kg of water vapour to condense it into liquid.

- c) Sarah thinks the water could be from her wet clothes that she left in her tent overnight.

- Explain the energy changes and direction the energy moves, as the water vapour **evaporates** from the wet clothes and then **condenses** onto the inside of the tent.

This image shows a full page of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page, providing a template for handwriting practice or general note-taking. There are no margins, text, or other markings on the page.

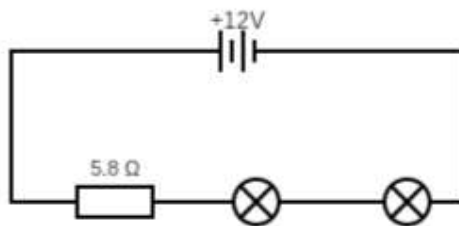
- d) If the air inside Sarah's tent is dry, calculate the amount of heat energy required to raise the temperature of 150 kg of dry air by 5°C .

- e) If the air in the tent was **moist** instead, explain how the energy required to raise it by 5°C would be different. Calculations are not required.

Question 3: **Electrical energy**

It started to rain which caused Sarah's campfire to go out. She wants to use her new food warmer instead to warm up some food, which can be plugged into her car.

She turns on her car so that the dashboard is lit up, the dashboard circuit is shown below:



- a) The circuit contains 2 identical bulbs ($3.4\ \Omega$ each) and 1 resistor. Calculate the total resistance of the circuit.

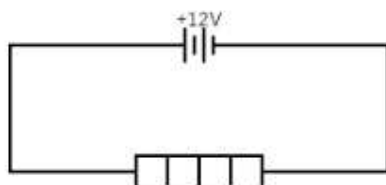
- b) Show current through the $5.8\ \Omega$ resistor is 0.95A .

c) Calculate

the power output of one of the lamps.

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- d) One of the lamps was replaced with energy efficient bulbs that have a lower power rating than the original bulb but the same brightness. Explain whether the energy-efficient light bulb has a higher or lower resistance than the original light.

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- e) When Sarah plugs in her food warmer, it consumes 27,000J of energy during a time of 15 minutes. It has one heating element as shown below:



- Calculate the resistance of **one** of the heating elements when the food warmer is switched on. Start by calculating the power produced by the heating element in 15 minutes.
